Probability and Statistics

Boxplot and Stem-Leaf Plot

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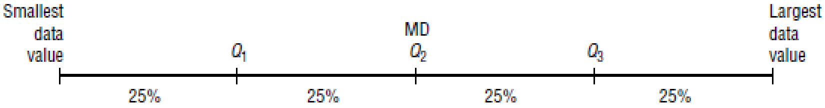
Percentiles divide the data set into 100 equal groups.

Quartiles divide the distribution into four groups, separated by Q_1, Q_2, Q_3.

Note that Q_1 is the same as the 25^{th} percentile; Q_2 is the same as the 50^{th} percentile, or the median; Q_3 corresponds to the 75^{th} percentile, as shown



Procedure Table

Finding Data Values Corresponding to Q_1, Q_2, and Q_3

Step 1 Arrange the data in order from lowest to highest.

Step 2 Find the median of the data values. This is the value for Q_2.

Step 3 Find the median of the data values that fall below Q_2. This is the value for Q_1.

Step 4 Find the median of the data values that fall above Q_2. This is the value for Q_3.

Find Q_1, Q_2, and Q_3 for the data set
15, 13, 6, 5, 12, 50, 22, 18.

Step 1 Arrange the data in order.

5, 6, 12, 13, 15, 18, 22, 50

Step 2 Find the median (Q_2).

$$\begin{array}{c} 5, 6, 12, 13, \; 15, 18, 22, 50 \\ \uparrow \\ \text{MD} \end{array}$$

$$\text{MD} = \frac{13 + 15}{2} = 14$$

Step 3 Find the median of the data values less than 14.

$$\begin{array}{c} 5, 6, \; 12, 13 \\ \uparrow \\ Q_1 \end{array}$$

$Q_1 = \dfrac{6 + 12}{2} = 9$ So Q_1 is 9.

Step 4 Find the median of the data values greater than 14

$$\begin{array}{c} 15, 18, \; 22, 50 \\ \uparrow \\ Q_3 \end{array}$$

$$Q_3 = \frac{18 + 22}{2} = 20$$

Here Q_3 is 20.

Hence, $Q_1 = 9$, $Q_2 = 14$, and $Q_3 = 20$

An **outlier** is an extremely high or an extremely low data value when compared with the rest of the data values.

An outlier can strongly affect the mean and standard deviation of a variable. For example, suppose a researcher mistakenly recorded an extremely high data value. This value would then make the mean and standard deviation of the variable much larger than they really were. Outliers can have an effect on other statistics as well. There are several ways to check a data set for outliers. One method is shown in this Procedure Table.

Check the following data set for outliers.
5, 6, 12, 13, 15, 18, 22, 50

The data value 50 is extremely suspect. These are the steps in checking for an outlier.

Step 1 Find Q_1 and Q_3. This was done in Example 3–36; Q_1 is 9 and Q_3 is 20.

Step 2 Find the interquartile range (IQR), which is $Q_3 - Q_1$.

$$\text{IQR} = Q_3 - Q_1 = 20 - 9 = 11$$

Step 3 Multiply this value by 1.5.

$$1.5(11) = 16.5$$

Step 4 Subtract the value obtained in step 3 from Q_1, and add the value obtained in step 3 to Q_3.

$$9 - 16.5 = -7.5 \qquad \text{and} \qquad 20 + 16.5 = 36.5$$

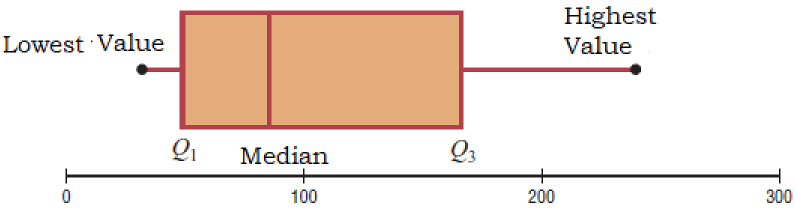
Step 5 Check the data set for any data values that fall outside the interval from -7.5 to 36.5. The value 50 is outside this interval; hence, it can be considered an outlier.

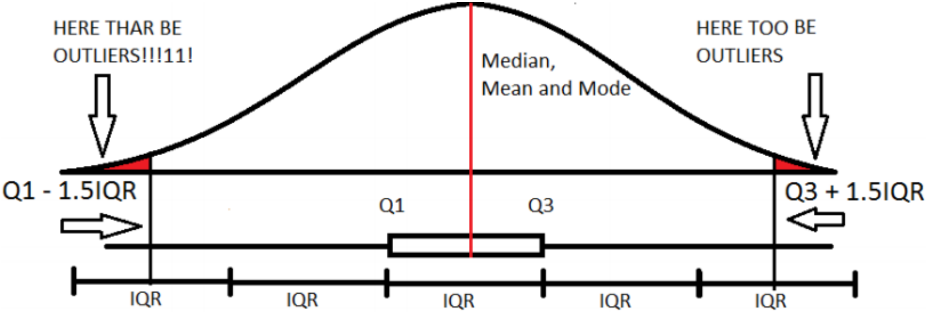
The Five-Number Summary and Boxplots

A boxplot can be used to graphically represent the data set. These plots involve five specific values:

1. The lowest value of the data set (i.e., minimum)
2. Q_1
3. The median
4. Q_3
5. The highest value of the data set (i.e., maximum) These values are called a five-number summary of the data set.

A **boxplot** is a graph of a data set obtained by drawing a horizontal line from the minimum data value to Q_1, drawing a horizontal line from Q_3 to the maximum data value, and drawing a box whose vertical sides pass through Q_1 and Q_3 with a vertical line inside the box passing through the median or Q_2.

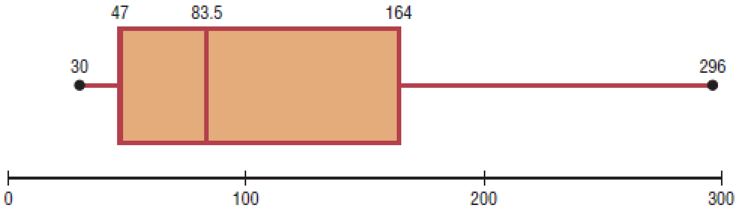



HERE THAR BE
OUTLIERS!!!11!
HERE TOO BE
OUTLIERS
Median,
Mean and Mode
Q1 - 1.5IQR
Q3 + 1.5IQR
Q1
Q3
IQR
IQR
IQR
IQR
IQR

The number of meteorites found in 10 states of the United States is 89, 47, 164, 296, 30, 215, 138, 78, 48, 39. Construct a boxplot for the data.

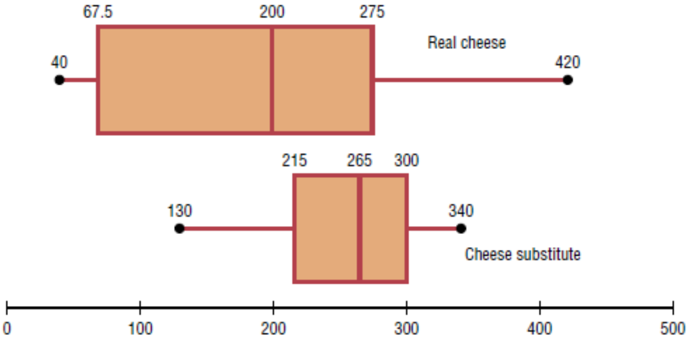
Q_1, Q_2 and Q_3 are $47, 83.57$ and 164, respectively.

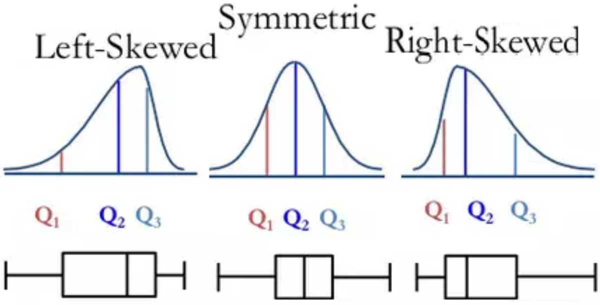
- Draw a scale for the data on the x axis.
- Located the lowest value, Q1, median, Q3, and the highest value on the scale.
- Draw a box around Q1 and Q3, draw a vertical line through the median, and connect the upper value and the lower value to the box.

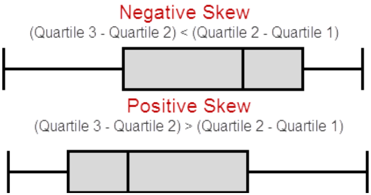


A dietitian is interested in comparing the sodium content of real cheese with the sodium content of a cheese substitute. The data for two random samples are shown. Compare the distributions, using boxplots.

Real cheese				Cheese substitute			
310	420	45	40	270	180	250	290
220	240	180	90	130	260	340	310




Left-Skewed
Symmetric
Right-Skewed
Q1 Q2 Q3
Q1 Q2 Q3
Q1 Q2 Q3


Negative Skew
(Quartile 3 - Quartile 2) < (Quartile 2 - Quartile 1)
Positive Skew
(Quartile 3 - Quartile 2) > (Quartile 2 - Quartile 1)

Symmetric Distribution When the data values are evenly distributed about the mean, a distribution is said to be a symmetric distribution. (A normal distribution is symmetric.)

Skewed Distribution When the majority of the data values fall to the left or right of the mean, the distribution is said to be skewed.

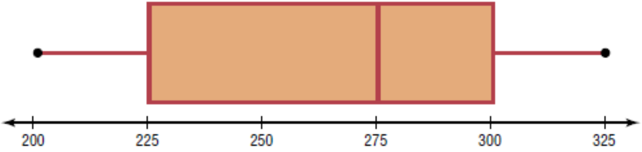
Negatively/left-skewed When the majority of the data values fall to the right of the mean, the distribution is said to be a negatively or left-skewed distribution.

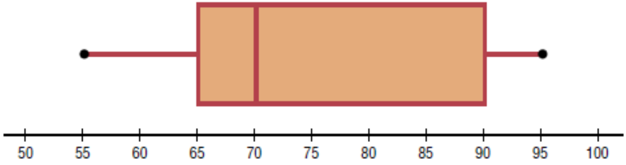
Positively/right-skewed When the majority of the data values fall to the left of the mean, a distribution is said to be a positively or right-skewed distribution. The mean falls to the right of the median, and both the mean and the median fall to the right of the mode.

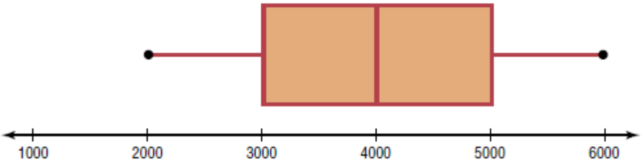
Information Obtained from a Boxplot

- If the median is near the center of the box, the distribution is approximately symmetric.
- If the median falls to the left of the center of the box, the distribution is positively skewed.
- If the median falls to the right of the center, the distribution is negatively skewed.

use each boxplot to identify the maximum value, minimum value, median, first quartile, third quartile, and interquartile range.







Construct a boxplot and comment on its skewness for the number of tornadoes recorded each month in 2005.

33, 10, 62, 132, 123, 316, 138, 123, 133, 18, 150, 26

A **stem and leaf plot** is a data plot that uses part of the data value as the stem and part of the data value as the leaf to form groups or classes.

At an outpatient testing center, the number of cardiograms performed each day for 20 days is shown. Construct a stem and leaf plot for the data.

25	31	20	32	13	14	43	36	32	33
32	44	02	57	32	52	44	51	45	23

1. Arrange the data in order:
2. Separate the data according to the first digit,
3. A display can be made by using the leading digit as the **stem** and the trailing digit as the **leaf**.

Stem and Leaf Plot for

Stem	Leaves
0	2
1	3 4
2	0 3 5
3	1 2 2 2 2 3 6
4	3 4 4 5
5	1 2 7

An insurance company researcher conducted a survey on the number of car thefts in a large city for a period of 30 days last summer. The raw data are shown. Construct a stem and leaf plot by using classes 50−54, 55−59, 60−64, 65−69, 70−74 and 75−79.

52	62	51	50	69
58	77	66	53	57
75	56	55	67	73
79	59	68	65	72
57	51	63	69	75
65	53	78	66	55

Stem and Leaf Plot for

5	0 1 1 2 3 3
5	5 5 6 7 7 8 9
6	2 3
6	5 5 6 6 7 8 9 9
7	2 3
7	5 5 7 8 9

When the data values are in the hundreds, such as 325, the stem is 32 and the leaf is 5. For example, the stem and leaf plot for the data values 325, 327, 330, 332, 335, 341, 345, and 347 looks like this.

32	5 7
33	0 2 5
34	1 5 7

The number of stories in two selected samples of tall buildings in Atlanta and Philadelphia is shown. Construct a back-to-back stem and leaf plot, and compare the distributions.

Atlanta					Philadelphia				
55	70	44	36	40	61	40	38	32	30
63	40	44	34	38	58	40	40	25	30
60	47	52	32	32	54	40	36	30	30
50	53	32	28	31	53	39	36	34	33
52	32	34	32	50	50	38	36	39	32
26	29								

Atlanta		Philadelphia
9 8 6	2	5
8 6 4 4 2 2 2 2 2 1	3	0 0 0 0 2 2 3 4 6 6 6 8 8 9 9
7 4 4 0 0	4	0 0 0 0
5 3 2 2 0 0	5	0 3 4 8
3 0	6	1
0	7	

Back-to-Back Stem and Leaf Plot